

# Innovative Deep Learning Approaches for ECG-Based Cardiovascular Disease Detection

Thirumala Narasimha Poluru, et al.

International Journal of Novel Research and Development (IJNRD), 2025

## ABSTRACT

Cardiovascular disease remains the leading cause of death worldwide, and early detection from electrocardiogram (ECG) signals is one of the most practical paths to reducing that toll. This work explores deep learning architectures for classifying cardiovascular conditions directly from ECG signals, comparing convolutional and recurrent approaches on preprocessed signal data. We address the practical problems that keep such models out of clinics - noisy real-world signals, class imbalance across rarer conditions, and the gap between benchmark accuracy and dependable behaviour on unseen patients - using signal preprocessing, augmentation, and careful evaluation protocols. The approach achieves strong classification performance while remaining computationally light enough for modest hardware.

## KEY CONTRIBUTIONS

1. A preprocessing and augmentation pipeline robust to noisy, real-world ECG signals.
2. A comparative evaluation of CNN and recurrent architectures for multi-class detection.
3. Class-imbalance handling so rarer conditions are not drowned out by common ones.
4. A deployment-conscious design keeping inference practical on modest hardware.

## KEYWORDS

Deep Learning, ECG Classification, Cardiovascular Disease, Signal Processing, ML in Healthcare

## NOTE

This file is a summary sheet generated for the portfolio. Replace it with the full published PDF at [frontend/public/research/paper.pdf](#) when convenient.